

Rajat Singh

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Professional Summary

AI Engineer specializing in LLMs, Agentic AI, NLP, and Generative AI. Experienced in building production-ready AI applications using LangChain, LangGraph, transformer models, RAG pipelines, and scalable cloud-native architectures.

Experience

AI/Cloud Intern, IBM SkillsBuild

Jul 2025 – Aug 2025

Developed intelligent AI applications using IBM Watson, Watson Assistant, and IBM Cloud. Built conversational AI systems with context management, intent classification, and entity recognition. Integrated AI models with enterprise workflows and cloud infrastructure.

Design Lead, GDG on Campus (Dyal Singh College)

Nov 2024 – June 2026

Key Projects

FrameForge (Manim Animation Generator)

[GitHub](#) | [Demo](#)

- Reduced animation generation failures by 90% via an 8-pass auto-fixer pipeline (LaTeX substitution, deprecated API patching, AST validation) that intercepts and repairs LLM-generated Manim code before render
- Cut average latency from 60s to under 2s for 9 high-frequency prompt types via a template cache with LLM intent verification, routing novel requests through a full Groq Llama 3.3 70B → validator → Supabase pipeline

Document Grounded QA System

[GitHub](#) | [Demo](#)

- Developed a RAG Q&A system processing 130+ page documents with 95% accuracy, <3s query latency, and 10,000+ chunk capacity using LangChain, FAISS vector store, and optimized chunking (1500/300 tokens) for 40% improved retrieval precision.
- Implemented custom prompt engineering reducing hallucinations by 60% and achieving 98% source attribution accuracy through retrieval-aware prompts and semantic search optimization.

Lockyt - Password Manager

[GitHub](#) | [Demo](#)

- Architected zero-knowledge encryption pipeline for Lockyt using AES-256-GCM client-side encryption and PBKDF2 key derivation (600,000 iterations), ensuring 0 bytes of plaintext are stored server-side across 5 vault item types.
- Built and shipped Lockyt's Chrome Extension (MV3) with sub-second autofill on any website, alongside a 4-dimension security center featuring breach detection, password strength analysis, reuse detection, and expiry alerts.

RAG Customer Support Chatbot

[GitHub](#) | [Demo](#)

- Architected a RAG-powered chatbot processing 27,000+ support queries with 90%+ accuracy using SentenceTransformer embeddings (all-MiniLM-L6-v2) and FAISS; implemented a hybrid response system with 0.80 similarity threshold, reducing LLM API calls by 60% while maintaining <2s latency.
- Optimized retrieval using 384-dimensional embeddings with logarithmic time complexity over 27,000+ vectors, achieving 95% context relevance with session-based chat history via Groq LLM (Gemma-2-9B-IT).

Education

2022 – Present BSc (Hons) Computer Science, **University of Delhi** (8.5 GPA)

2022 Class 12th (95.2%)

2020 Class 10th (94.6%)

Technical Skills

AI & LLM Frameworks: LangChain, LangGraph, Hugging Face Transformers, SentenceTransformers, Groq API, Scikit-learn, TensorFlow, PyTorch, FAISS

Programming Languages: Python, SQL, JavaScript, C++

Backend & APIs: FastAPI, Flask, REST APIs, React

Cloud, Databases & DevOps: AWS EC2, AWS S3, MongoDB, Supabase, Docker, Kubernetes, Git, Linux

Tools: Streamlit, MLflow

Certificates

2025 Mastering Generative AI and ChatGPT, GeeksforGeeks

2025 TensorFlow Developer Certification, GeeksforGeeks

2024 Google Cloud Platform Essentials

Achievements

- Solved **350+** Data Structures and Algorithms problems on [Leetcode](#)
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